Chapter 2
Knowing Metrology and Its Structure



2.1 Metrology: Introduction

Metrology supports a universal agreement for units of measure, that is, the standardization of values. For this to happen, there must be an international and national metrological structure to ensure that the measuring instruments are maintained and applied properly and correctly in daily operational and business transactions. This standardization of units of measure is of great commercial importance for nations and companies.

For example, car manufacturing has several parts suppliers, each with its own production system and measurement instruments. However, all parts should fit perfectly into the car assembly. Imagine a wheeled supplier manufacturing and measuring the holes of the fixing screws with a slightly smaller diameter than those manufactured and measured by the screw supplier; the wheels could not be used.

As we saw in the previous chapter, using different units of measurement for the same quantity conflicts with the standardization of language established in the International System of Units (SI). However, some British colonization countries still employ other units of measure, such as the inch, foot, pound, yard, and mile.

This use can lead to severe misconceptions and disastrous consequences for society. Let us consider some real cases as examples.

Case 2.1: Vasa, the Swedish Warship (Fig. 2.1)

Swedish warship Vasa wrecked in 1628 on its inaugural trip, less than two kilometers from the coast, causing the death of 30 crew members. At the time, armed with 64 bronze cannons, it was considered the most powerful ship in the world. The archeologists who studied him after he was lifted from the bottom of the sea in 1961 said he was thicker to the stubborn than the stubby. One reason may be that the workers used different systems of measurements, as archeologists found four

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P. P. Novellino do Rosario, A. Mendes, *Metrology and Measurement Uncertainty*,
https://doi.org/10.1007/978-3-031-82303-9_2

2.2 Importance of Measuring

Measuring is part of our daily lives:

- When looking at the display of a clock, we see the result of the time measurement (hour, minute, and second);
- When buying a heavy product on a scale, we have the mass measurement (kilogram, gram);
- When supplying the car at the gas station, we noticed the (liter) volume measurement of fuel;
- When we received the electricity bill from our residence, we could see the electricity consumption measured in that period (kW).

Anyway, we are constantly witnessing and experiencing the results of measurements.

Measuring is a process that involves the existence of:

- A phenomenon (either a quantity or a substance) we want to know;
- A measuring instrument (or a set of instruments) calibrated, preferably;
- A unit of measure (kg, m, °C, etc.);
- An individual trained to perform the act of measuring and correctly interpreting the result.

Two great scientists highlighted the importance of measuring many years ago (Figs. 2.5 and 2.6).

> Measure what is measurable, and make measurable what is not.
> (https://mathshistory.st-andrews.ac.uk/Biographies/Galileo/quotations/)

> When you measure what you are speaking about and express it in numbers, you know something about it, but when you cannot, your knowledge about it is meager and unsatisfactory.
> (https://mathshistory.st-andrews.ac.uk/Biographies/Thomson/quotations/)



Fig. 2.5 Galileo Galilei. (https://tse4.mm.bing.net/th?id=OIP.5WKAICf1vtt9p2OtlS3wNQHaHa&pid=Api&P=0&h=180)

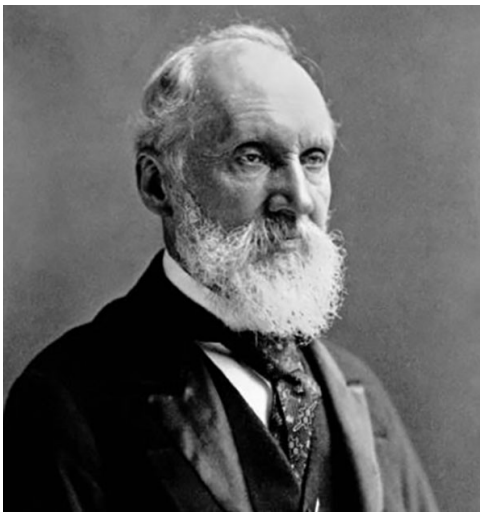


Fig. 2.6 William Thomson (Lord Kelvin). (https://res.cloudinary.com/dk-find-out/image/upload/q_80,w_1920,f_auto/A-Corbis-IH190312_l5pagc.jpg)

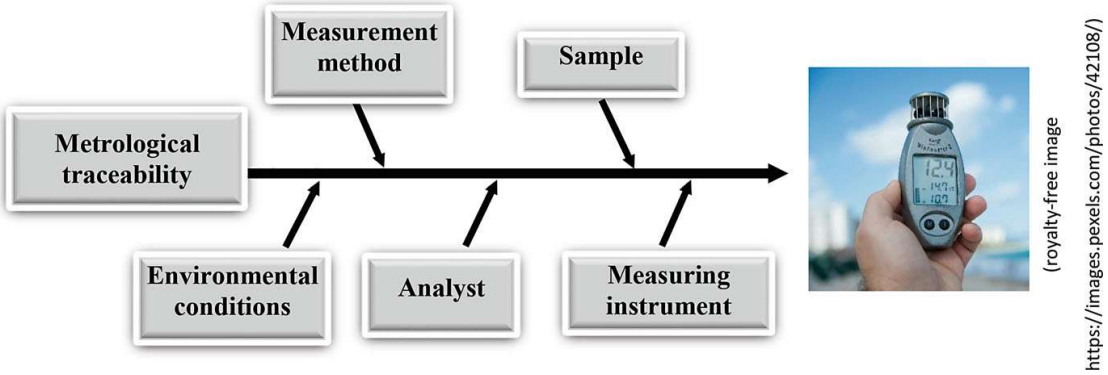


Fig. 2.7 Metrological agents

2.3 Measurement Objective

Decisions must be made based on information in any field of activity. In the scientific and technological area, such information is generally the result of measurements directly or indirectly related to the object under study.

Measurement is the "process of experimentally obtaining one or more quantity values that can reasonably be attributed to a quantity."

Measurements can be influenced by different metrological agents, such as (i) the measurement method, (ii) the sample, (iii) the analyst, (iv) the measurement instrument, (v) the environmental conditions, and (vi) the traceability of the measurement instruments and standards. Thus, we understand the measure as the "result of the measurement process," in this sense, its quality depends on how such a process is managed.

Figure 2.7 presents the different metrological agents that influence the measurement result.

Let us discuss each metrological agent briefly.

2.3.1 Measurement Method

Here is a definition of measurement method: "*generic description of a logical organization of operations used in a measurement*" [VIM—2.5].

The measurement method should ideally be contained in a technical standard; however, it may also be present in an operational procedure, a work instruction, a flowchart, or any other internal organization document.

It should be noted that the measurement method should be developed by experts in the subject and used by professionals with knowledge and training in the techniques defined by the process.

Example 2.1
The standard CEN-EN 837-1 Pressure Gauges—Part 1: Bourdon Tube Pressure Gauges—Dimensions, Metrology, Requirements, and Testing determines the conditions for calibrating the Bourdon-type gauge. Among the various requirements, it defines, for example, the minimum number of calibration points as a function of the accuracy class (Table 2.1).

Accuracy class: Class of measuring instruments or measuring systems that meet stated metrological requirements that are intended to keep measurement errors or instrumental measurement uncertainties within specified limits under specified operating conditions.

NOTE 1 An accuracy class is usually denoted by a number or symbol adopted by convention.
NOTE 2 Accuracy class applies to material measures. [VIM—4.25].

Knowing a Little More... (Fig. 2.8)

Eugène Bourdon was born in 1808 in Paris, France. He began his career as a watchmaker and later as an engineer. In 1849, Bourdon invented the meter that bears his name. This gauge can measure up to 6800 atmospheres. This invention also helped decrease the number of steam engines because, before Bourdon, measuring this amount of pressure was almost impossible. He died in 1884, but his Bourdon tube pressure gauge is still used today.

Table 2.1 Number of calibration points of a pressure gauge according to CEN-EN 837-1

Accuracy class	Number of calibration points
0.1; 0.25; 0.6	10
1; 1.6; 2.5	5



Fig. 2.8 Eugène Bourdon. (Photo: public domain)



Fig. 2.9 Sample of screws to determine the thread step. (https://images.pexels.com/photos/259988/)

2.3.2 Sample

A sample is a representative part of a whole that allows the results to be attributed to the original set once they have been evaluated, analyzed, and measured.

Example 2.2
Let us check the thread step of a lot of 10,000 screws (Fig. 2.9).

One option would be to measure all 10,000 screws. However, it would be a time-consuming and costly process. Thus, a viable alternative is to randomly choose a certain number of production parts as a sample, using criteria established in a technical standard, for example. The average step measurements of the selected pieces are considered a reasonable estimate for the total screws.

We must be careful when selecting and using a sample to represent the set; otherwise, we can assign wrong values due to improper choice or handling of the sample.

Some primary care should be observed in the choice and definition of the sample:

- Apply statistical methods to determine the sample size since it should represent the whole;
- Make the random selection of the sample and ensure that it belongs to the same manufacturing batch. For example, an excellent way to determine the ambient temperature of a laboratory is to measure temperature in various locations, not just one place;
- Ensure that measurements are performed under conditions defined in standards, methods, or technical procedures. Example: using CEN-EN 837-1 again, it represents that the temperature of the calibration site must be comprised between (20 ± 2) °C;
- Avoid contamination that may modify the physical or chemical characteristics of the sample;
- Check, where applicable, the validity of the sample.

2.3.3 Analyst

The analyst, human factor, and central element of the measurement process (Fig. 2.10) need:

- Know the measurement method;
- Know how to evaluate environmental conditions and decide on whether or not to measure measurements;
- Be able to select the sample to be evaluated adequately;
- Be trained for the correct use of the instruments that make up the measurement system;
- Register and correctly interpret the result of the measurements.



Fig. 2.10 Ultrasonography examination measurement system. (https://images.pexels.com/photos/7089400/)

2.3.4 Environmental Conditions

Initially, we need to define *influence quantity*:

> *Quantity that, in a direct measurement, does not affect the quantity that is measured, but affects the relation between the indication and the measurement result.*
>
> *Example: Temperature of a micrometer used for measuring the length of a rod, but not the temperature of the rod itself, which can enter into the definition of the measurand.* [VIM—2.52].

Influence quantities can usually not be avoided but should be monitored and controlled to minimize their effects on the measurement result.

Thus, what we call environmental conditions are the influences of these environmental factors, such as temperature, humidity, dust, vibration, fluctuation in the power supply, electric or magnetic noise, lighting, or other factors in a place where measurements will be realized.



Fig. 2.11 Digital thermohygrometer (temperature and relative humidity). (https://pixabay.com/photos/time-clock-humidity-air-hygrometer-2353382/)

Example 2.3
To measure the concentration of a particular active ingredient that enters a medicine's composition, the laboratory temperature must be maintained at (22.0 ± 0.5) °C and relative humidity at (50 ± 5) %.

Under ideal conditions, an air conditioner system should control temperature and moisture.

A thermohygrometer (Fig. 2.11) must measure these conditions to enable the analyst to take action if these variables leave control.

We must interrupt measurements or correct their results when any anomaly arises, either in temperature or moisture (Fig. 2.11).

2.3.5 Measuring Instrument

A measuring instrument is a *"device used for making measurements, alone or in conjunction with one or more supplementary devices."* [VIM—3.1].

Examples 2.4
Some measuring instruments (Figs. 2.12, 2.13, 2.14, 2.15, and 2.16).

Measurement indicates or controls a process, monitors an alarm, or investigates a physical, chemical, or biological phenomenon. In simple monitoring, measurement

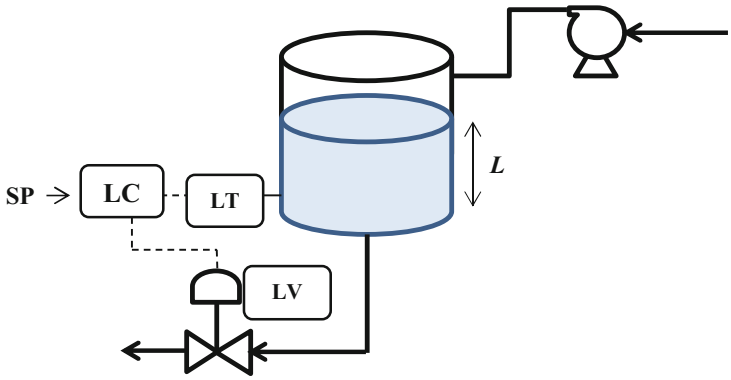


Fig. 2.18 Level control system

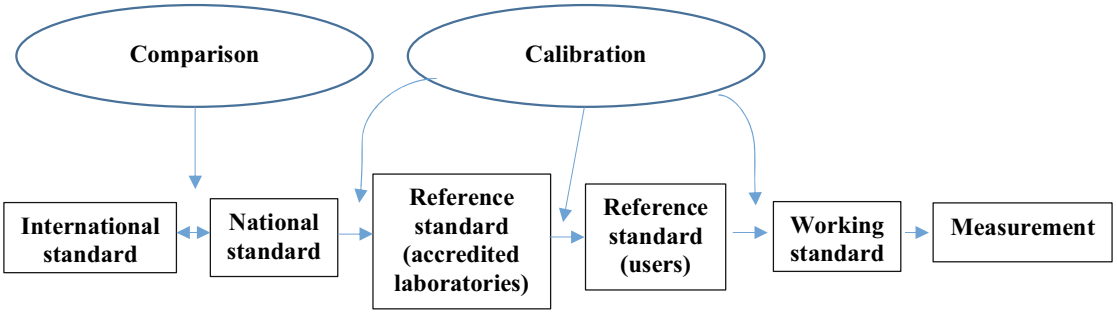


Fig. 2.19 Metrological traceability

Knowing a Little More...
See the Level control system of a reservoir, as shown in Fig. 2.18.

The transducer (LT) sends to the controller (LC) an electrical signal proportional to the level (L) variation in the water tank. The controller compares this sign of the process variable (level) to a reference (set point) value (SP) and, depending on the magnitude of this difference, sends a correction signal to the control valve (LV), so that it reduces (or increases) the liquid flow rate to keep the level stable within the reservoir.

An alarm system operates on warning, sound, or visual warning equipment after an unwanted or dangerous situation (e.g., a fire alarm).

The alarm system can also operate with safety systems to maintain equipment integrity, especially for people.

When investigating a phenomenon, an example is measuring the "hole in the ozone layer" in the earthly atmosphere to determine its consequences for life on the planet.

2.3.6 Metrological Traceability

The definition of *metrological traceability* is (Fig. 2.19):

> *Property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty*. [VIM—2.41].

The calibrated standards and instruments with guaranteed traceability transfer accuracy to measurements, enabling an adequate estimate of the final measurement uncertainty. Thus, before performing and using a measurement's result as relevant information for any decision, it is necessary to analyze the measurement process to know all sources of influence associated with metrological agents.

Once these sources of influence are identified, the measurement process must be active to yield quality measures, that is, metrological reliability.

Thus, the uncertainty derived from each metrological agent influences the final uncertainty of the measurement process. In the measurement chapters (Chaps. 5 and 6), we will address the concept of uncertainty and the methodology for its estimate, considering the variables of influence.

2.4 Metrological Reliability

Generically, reliability is the capacity or probability of a system to perform a function and maintain its operation under specific conditions, correctly, as provided in the project, during a predetermined period, under routine circumstances, as well as in hostile and unexpected circumstances.

Thus, metrological reliability is the ability of a measurement system to convey certainty and confidence in the results obtained. Without metrological proof, there is no way to guarantee the reliability of control data of characteristics that determine the quality of the product.

Analyzing the environment on the consumer side, the existing metrological system should enable users' access to compliance verification mechanisms of the products offered. From the results of the measurements performed by manufacturers and verified by the controlling agencies, consumers may trust that industrialized products have been adequately measured (e.g., weight, volume, chemical composition, concentration, etc.) and released for commercialization.

2.5 Metrology Areas of Expertise

We can separate metrology into two significant areas of activity: Legal metrology and Scientific and industrial metrology.

2.5.1 Legal Metrology

It is the area of metrology closest to the ordinary citizen. Its primary function is to protect products and services that involve and need measurement.

The International Organization of Legal Metrology (OIML) defines it as "the application of legal requirements for measures and measuring instruments."

Metrological regulations based on the OIML guidelines establish technical requirements, metrological control, use, and marking requirements, and the units of measure that measuring instrument manufacturers and users must meet.

In addition to commercial activities, measuring instruments used in official activities, in the medical area, in the manufacture of medicines, and the fields of occupational, environmental, and radiation protection are subjected to metrological control. In these cases, control assumes particular importance in the face of the dangerous adverse effects of wrong results on human health.

Table 2.2 shows the measuring instrument categories included in the OIML Certification System and the corresponding OIML Recommendations.

The OIML Certification System (OIML-CS) is a system for issuing, registering, and using OIML certificates and their associated OIML-type evaluation/test reports for types of measuring instruments (including families of measuring instruments, modules, or families of modules) based on the requirements of OIML Recommendations.

2.5.2 Scientific and Industrial Metrology

Scientific metrology is linked to scientific research and methodologies of the highest metrological quality. It deals with measurement standards and laboratory instruments.

As they unfold, these actions also include industry measurement systems (industrial metrology), which control production processes and ensure the quality of products and services offered to the market.

2.6 International Metrological Structure

The international structure of each of the two significant areas of Metrology (legal and scientific) is very similar.

2.6.1 Legal Metrology

International Organization of Legal Metrology (OIML)
It is an intergovernmental treaty that, among other activities, develops regulations, rules, and documents for use by the legal and industry authorities.

Fig. 2.20 OIML



Knowing a Little More... (Fig. 2.20)

Since 1955, OIML has launched the foundations for a world metrology system.

The mission of the OIML is to enable economies to put in place effective legal metrology infrastructures that are mutually compatible and internationally recognized, for all areas for which governments take responsibility, such as those which facilitate trade, establish confidence and harmonize the level of consumer protection worldwide.

Available in: https://www.oiml.org/en

International Conference of Legal Metrology
OIML's maximum decision-making body. The conference, which takes place every four years, comprises representatives of member countries, countries that come together as observers, and associations of international institutions. Its purpose is to define the general policy and promote the implementation of OIML metrological guidelines.

International Committee on Legal Metrology (CIML)
It is the organization's functional decision body. Approve the annual BIML work plan and adopt OIML recommendations, documents, and publications.

International Bureau of Legal Metrology
It is OIML's secretariat and headquarters. The bureau organizes conference and commission meetings, executes conference decisions and commissions, and disseminates and distributes the organization's publications (Fig. 2.21).

2.6.2 Scientific Metrology

General Conference of Weights and Measures (CGPM)
The CGPM is made up of member state delegates and associate observers. Among its attributions is the discussion and analysis of the necessary provisions to ensure the propagation and improvement of the SI.



Fig. 2.21 BIML. (https://www.oiml.org/en/ressources/biml/sacre_coeur.jpg)

Fig. 2.22 BIPM

International Committee on Weights and Measures (CIPM)
Composed of 18 member countries, it acts as an international scientific authority, and its main task is to promote world uniformity in units of measure through direct action or the presentation of resolution projects to the CGPM.

International Bureau of Weights and Measures (BIPM) (Fig. 2.22)

An intergovernmental organization established by the Meter Convention in 1875 aims to ensure and promote the global comparability of measurements, including the supply of an international unit system (SI) and the International Reference Time Scale (UTC) for scientific research and innovation.

2.7 Regional Metrological Structure

2.7.1 EURAMET—The European Association of National Institutes of Metrology

The mission is to develop and disseminate an integrated, profitable, and competitive measurement infrastructure for Europe, always considering the needs of the industry, companies, and governments. With its services, EURAMET supports members in meeting national requirements and establishing a balanced European measurement

international leadership and recognition, and foster collaboration throughout the scientific community of the measurement.

(Text adapted from https://www.euramet.org)

2.7.2 *Inter-American Metrology System (SIM)*

The Inter-American Metrology System (SIM), instituted in 1979, resulted from a broad agreement between the national metrology organizations involving 34 nations. Its mission is to promote and support an integrated measurement infrastructure in the Americas, which allows each National Institute of Measurement to encourage innovation, competitiveness, trade, consumer security, and sustainable development, effectively participating in the international metrology community.

Organized in five subregions (Noramet, Carimet, Camet, Andimet, and Suramet), it has a governor board structured by a coordinator of each subregion, a technical committee, a professional development committee, and an integrated representation that provides access to SIM in a worldwide agreement to compare standards at the highest metrology level.

SIM is committed to implementing a global measurement system in the Americas that will ensure the confidence of all users. Working to establish a robust regional measuring system, SIM is essential to the development of a free trade area in the Americas.

In the context of established cooperation, measures taken by the member countries will help to achieve:

- Establishment of national and regional measurement systems.
- Establishment of a hierarchy of the national standards of each country and binding on regional and international standards.
- Establishment of equivalence between national measurement standards and calibration certificates issued by national metrology laboratories.
- Comparability of the results obtained in measurement processes performed in laboratories within the system.
- Training of technical and scientific personnel.
- Distribution of technical and scientific documentation.
- Binding with international standards maintained by the International Bureau of Weights and Measures (BIPM).
- Straight cooperation with the BIPM, the OIML, and other international organizations interested in laboratory accreditation (ILAC) and with technology and measurement patterns (IMEKO), research and development (universities and organizations P&D), oriented to promote competitiveness, promote more equitable business transactions and support essential development in health, safety, sustainable industrial development, and environmental protection.

(Text adapted from https://sim-metrologia.org/)

2.7.3 Other Regional Metrological Structures

The Asia Pacific Metrology Programme (APMP)
The APMP is a grouping of national metrology institutes (NMIs) from the Asia-Pacific region engaged in improving regional metrological capability by sharing expertise and exchanging technical services among Member laboratories. The APMP is one of the six Regional Metrology Organizations (RMOs) that implement the CIPM MRA for the worldwide mutual recognition of measurement standards and calibration and measurement certificates. It is also one of the Specialist Regional Bodies (SRBs) working with Asia-Pacific Economic Cooperation (APEC) to facilitate developing and implementing standards and conformance infrastructures that address APEC goals.

(Text adapted from https://apmpweb.org/)

Euro-Asian Cooperation of National Metrological Institutions (COOMET)
COOMET is an organization for the Euro-Asian cooperation of National Metrology Institutions (from the countries of Central and Eastern Europe, Asia, and nearby countries). It is open to the National Metrology Institutions of countries from other regions to join it. Its mission is to raise the level of metrology development, support, and expansion of an integrated measurement infrastructure for countries in the Euro-Asian region and other interested countries that makes it possible for every national metrological institute to promote innovation, competitiveness, trade, consumer safety, sustainability and ensuring international recognition. Main areas of cooperation: Measurement standards of physical quantities; Legal metrology; Quality management systems; Information and training; Innovative research in metrology.

(Text adapted from https://www.coomet.net/)

Southern African Development Community Cooperation in Measurement Traceability (SADCMET)
The SADC Cooperation in Measurement Traceability coordinates metrology activities and services in the region to provide regional calibration and testing services, including regulatory bodies, with readily available traceability to the SI units of measurement through legally defined and regionally and internationally recognized national measurement standards. Its primary objectives are to (i) Promote closer collaboration among its members in their work on measurement standards within the present decentralized regional metrology structure; (ii) Improve existing national measurement standards and facilities and make them accessible to all members; (iii) Ensure that new national measurement standards and facilities developed in the context of SADCMET collaborations are accessible to all members; (iv) Contribute to the formulation of and participate in intra- and interregional systems to maintain the continued traceability of the national measurement standards of the SADC member states to the SI units of measurement; (v) Encourage the harmonization of legislation relating to national measurement standards.

(Text adapted from http://www.sadcmet.org/SitePages/Home.aspx)

2.8 Metrological Structure in Brazil

2.8.1 National System of Metrology, Standardization, and Industrial Quality (SINMETRO)

SINMETRO was instituted by Law Number 5966 of December 11, 1973, and is tasked with managing the technological services infrastructure in metrology (legal, scientific, and industrial), standardization, industrial quality, and conformity assessment.

Compose the SINMETRO:

- National Council of Metrology, Standardization and Industrial Quality (CONMETRO) and its technical committees.
- National Institute of Metrology, Quality and Technology (INMETRO).
- Brazilian Association of Technical Standards (ABNT).
- Certification bodies for quality systems, environmental management, products, and personnel; Inspection organisms; Training bodies; Proficiency testing bodies.
- Accredited calibration and testing laboratories.
- State Institutes of Weights and Measures (IPEM) and.
- State metrological networks.

SINMETRO Areas

Legal Metrology

The activities of Legal Metrology in Brazil are before the law that instituted the SINMETRO. In the 1930s, there was already a "Metrology Law," and metrological control began, in fact, with the creation of the National Institute of Weights and Measures (INPM) in 1961, replaced in 1973 by INMETRO, which incorporated its activities.

As stated earlier, legal metrology is one of the largest consumer protection systems. INMETRO coordinates the Brazilian Network of Legal Metrology and Quality (RBMLQ-I), which comprises the States' Weight and Measures Institutes (IPEM).

Scientific and Industrial Metrology

Scientific and industrial metrology promotes competitiveness and stimulates an environment favorable to the country's scientific and industrial development. It is also essential to technological innovation. INMETRO coordinates this process, is responsible for the fundamental metrological quantities with reliability equal to that of the countries of the first world, and transfers measurement standards to the society.

Testing and Calibrations

Responsibility for test activities (used for product certification) and calibrations (from standards and industrial instruments) within the SINMETRO are the laboratories that make up the Brazilian Testing Laboratories Network (RBLE) and the Brazilian Calibration Network (RBC). They are laboratories accredited by INMETRO and can be public, private, mixed, national, or foreign.

Uruguay Institute of Technical Standards (UNIT), and the National Institute of Technology, Standardization and Metrology (INTN—Paraguay).

International Accreditation Forum (IAF) (www.iaf.nu)

The World Association of Accreditation Bodies and other bodies are interested in conformity assessment in management systems, products, services, personnel, and similar conformity assessment programs. Its principal function is to develop a single worldwide conformity assessment program to reduce risk for companies and their customers and ensure that accredited certificates can be reliable.

Inter-American Accreditation Cooperation (IAAC) (www.iaac.org.mx)

It is an association whose mission is to promote cooperation between accreditation bodies and stakeholders of the Americas, aiming at developing conformity assessment structures to improve products, processes, and services. It was created in 1996 in Uruguay and incorporated in 2001 as a civil association according to Mexican law. It is not for profit and works based on the cooperation of its members and stakeholders. It obtains adhesion fees, voluntary contributions from its members, and donations based on regional organizations projects, particularly the Organization of American States (OAS) and the Physikalisch Technische Bundesanstalt (PTB) from Germany.

2.8.2 *Brazilian Laboratory Structure*

National Institute of Metrology, Standardization and Technology (INMETRO)
Their skills and attributions in the area of metrology are:

- Ensuring the standardization, maintenance, and dissemination of fundamental units of the international system (SI).
- Tracking the measurement units to international standards and spreading them to industries.
- Establishing the methodologies for the comparison of measurement standards, instruments, and materialized measures.
- Tracking the reference standards of the laboratories accredited to national standards.
- Acting in the area of legal metrology and supporting standardization and industrial quality activities.
- Accrediting laboratories and establishing value ranges and measurement uncertainty.

INMETRO's laboratories are home to technical divisions in acoustics and vibrations, electricity, mechanics, optics, thermal, and chemistry. They are responsible for:

(Text adapted from http://pcdsh01.on.br).

Laboratories Accredited by INMETRO

INMETRO grants accreditation based on the standard ISO/IEC 17025: 2017, according to the guidelines established by the International Laboratory Accreditation Cooperation (ILAC) and the Good Practices Codes (GPC) from the Organization for Economic Co-Operation and Development (OECD).

Accreditation is allowed to any laboratory that provides calibration or testing service, independently or linked to an organization, public or private, national or foreign, despite its size or area of expertise.

2.9 Technical Standards and Metrology

A technical standard establishes quality, performance, and safety requirements for providing something, its use, or its final destination. It also stipulates procedures, standardizes shapes, dimensions, types, and uses, fixes classifications or terminologies and glossaries, and defines how to measure and determine characteristics, such as test methods.

Technical standards apply to products, services, processes, and management systems in the most diverse fields. In general, the customer establishes the technical standard to supply the good or service he wants to acquire. This can be done explicitly when the customer clearly defines the applicable standard or expects that the rules in use will be followed in the market where it operates.

Important
We can say that there is no metrology without technical standards!

2.9.1 The ISO 9001:2015 and the Metrology

ISO 9001:2015—Quality Management Systems—Requirements [11] specifies requirements for a management system that can be used for internal application by organizations, certification, or contractual purposes. Focusing on the metrological issue, there is a specific technical requirement in the standard, *7.1.5.2 Measurement Traceability*, which establishes the following:

> *When measurement traceability is a requirement or is considered by the organization to be an essential part of providing confidence in the validity of measurement results, measuring equipment shall be:*
>
> (a) *calibrated or verified, or both, at specified intervals, or before use, against measurement standards traceable to international or national measurement standards; when no such*

2.9.2 The ISO/IEC 17025:2017 and the Metrology

The ISO/IEC 17025:2017 General requirements for the competence of testing and calibration laboratories [12] comprise testing and calibrations performed using standardized methods, non-normalized methods, and methods developed by the laboratory. Figure 2.23 summarizes the various normative requirements in which metrology is strongly present.

Facilities and environmental conditions—req. 6.3

- Monitor, control, and register environmental conditions
- Facilities and environmental conditions cannot adversely affect the validity of the results

Equipment—req. 6.4

- Laboratory must have all measuring instruments, standards, and reference materials required to perform their activities.
- Equipment capable of achieving the accuracy and measurement uncertainty required.
- Equipment must be calibrated.
- The laboratory must have a calibration program.
- Indicate calibration status.
- Ensure operation and calibration of an instrument that has come out of direct control of the laboratory.
- Intermediate checks performed according to procedure.
- Instruments protected against adjustments that invalidate results.

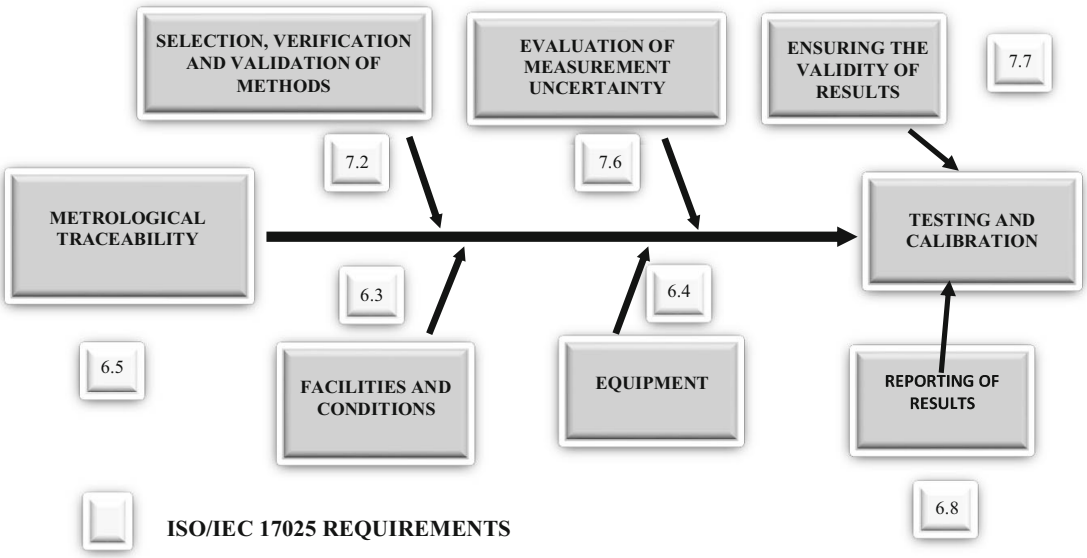


Fig. 2.23 Metrology in the ISO/IEC 17025:2017 requirements

Metrological traceability—req. 6.5

- Laboratory should establish and maintain metrological traceability of their measurement results.
- Calibrations and measurements traceable to the SI.
- Program and procedure for calibrating reference patterns by traceable organisms; Used only for calibration; calibrated before and after adjustments.
- Reference materials: traceable to SI units or already certified reference materials.

Selection, verification, and validation of methods—req. 7.2

- Use appropriate methods and procedures to evaluate the measurement uncertainty.
- Method validation includes, among other techniques, the calibration or evaluation of the trend and accuracy using standards or reference materials.

Evaluation of measurement uncertainty—req. 7.6

- Identification of contribution sources for measurement uncertainties.
- Calibration: assessment of measurement uncertainty for all calibrations.
- Testing: assessment of measurement uncertainty or a method based on the method.

Ensuring the validity of results—req. 7.7

- Use of certified reference materials.
- Interlaboratory comparison program or proficiency tests.
- Intermediate checks on measurement equipment.
- Replicated testing or calibrations.
- Retesting or recalibration of retained items.

Reporting of results—req. 7.8

We fully dedicate Chap. 9 of this book to discuss the importance of this requirement.

2.9.3 Laboratory Accreditation

Calibration Laboratories
The authorized laboratories to perform calibration services gather technical skills and abilities linked to industries, universities, and technological institutes and adopt standards traceable to national or international metrological references, establishing a relationship with the units of the International System of Units (SI).

Testing Laboratories
Like calibration, these laboratories gather skills and technical capacities associated with industries, universities, and technological institutes. They are trained to carry

out tests and performance tests on products with mandatory or voluntary certification. The traceability of the measures is guaranteed by calibration of standards in proven laboratories or directly in the laboratories of the National Metrology Laboratory.

2.9.4 *ILAC—International Laboratory Accreditation Cooperation (www.ilac.org)*

ILAC is the global association for the accreditation of laboratories, inspection bodies, proficiency testing providers, and reference material producers, with a membership consisting of accreditation bodies and stakeholder organizations worldwide. It is a representative organization that is involved with:

- *the development of accreditation practices and procedures,*
- *the promotion of accreditation as a trade facilitation tool,*
- *supporting the provision of local and national services,*
- *the assistance in developing accreditation systems,*
- *the recognition of competent testing (including medical) and calibration laboratories, inspection bodies, proficiency testing providers, and reference material producers around the world.*

ILAC actively cooperates with other relevant international organizations to pursue these aims. ILAC facilitates trade and supports regulators by operating a worldwide mutual recognition arrangement—the ILAC Arrangement—among Accreditation Bodies (ABs). The data and test results issued by laboratories and inspection bodies, collectively known as Conformity Assessment Bodies (CABs), accredited by ILAC Accreditation Body members are accepted globally via *this Arrangement. Thereby, technical barriers to trade, such as the re-testing of products each time they enter a new economy, are reduced to realize the free-trade goal of "accredited once, accepted everywhere." In addition, accreditation reduces risk for business and its customers by assuring accredited CABs are competent to carry out their work within their scope of accreditation. Further, the results from accredited facilities are used extensively by regulators for the public benefit in providing services that promote an unpolluted environment, safe food, clean water, energy, health, and social care services.* (Text obtained from ILAC-P14:09/2020—ILAC Policy for Measurement Uncertainty in Calibration)

ILAC first started as a conference, held on 24-28 October 1977 in Copenhagen, Denmark, to develop international cooperation for facilitating trade by promoting the acceptance of accredited test and calibration results. In 1996, ILAC became a formal cooperation with a charter to establish a network of mutual recognition agreements among accreditation bodies. In 2000, the 36 ILAC Full Members, consisting of laboratory accreditation bodies from 28 economies worldwide, signed the ILAC Mutual Recognition Arrangement (ILAC MRA) in Washington, DC, to promote the acceptance of technical test and calibration data for exported goods. The ILAC MRA for calibration and testing laboratories came into effect on 31 January 2001. The ILAC MRA was then extended in October 2012 to include the accreditation of inspection bodies. In May 2019, it was further extended to include the accreditation of proficiency testing providers and in May 2020 for the accreditation of reference material producers. (Text obtained from https://ilac.org/about-ilac/).

The ILAC Mutual Recognition Arrangement (ILAC MRA) provides the significant technical underpinning to the calibration, testing, medical testing, and inspection results, provision of proficiency testing programs and production of the reference materials of the accredited

conformity assessment bodies that in turn delivers confidence in the acceptance of services and results. The ILAC MRA supports the provision of local or national services, such as providing safe food and clean drinking water, providing energy, delivering health and social care, or maintaining an unpolluted environment. In addition, the ILAC MRA enhances the acceptance of products across national borders. Technical trade barriers are reduced by removing the need for additional calibration, testing, medical testing, and inspection of imports and exports. In this way, the ILAC MRA promotes international trade, and the free-trade goal of "accredited once, accepted everywhere" can be realized. (Text obtained from https://ilac.org/ilac-mra-and-signatories/).

On the ILAC website, you can search (https://ilac.org/signatory-search/) for accreditation bodies of the various countries to verify the activities of calibration and testing (ISO/IEC 17025), medical testing (ISO 15189), inspection (ISO/IEC 17020), proficiency testing providers (ISO/IEC 17043), and reference material producers (ISO 17034). Use this directory to find an accreditation body in the economy where you require the calibrations, testing, or inspections to be carried out.

2.9.5 The ISO 10012:2003 and the Metrology

The ISO 10012:2003 Measurement management systems—Requirements for measurement processes and measuring equipment[1] [13]—provide guidelines for measuring process management and metrological evidence of measuring instruments used to support and demonstrate conformity with metrological requirements.

The standard declares that an effective management system ensures the instruments and measurement processes are suitable for their intended use. It also points out that the management system should manage the risk that these measuring instruments and methods can produce incorrect results that affect the quality of an organization's products.

ISO 10012: 2003 is an "essentially metrological" standard, and all its requirements deal with important subjects. However, we will highlight requirement 7—metrological confirmation and realization of measurement processes—which presents a series of exciting guidelines, some of which we present to follow.

Knowing a Little More...
The following scheme represents a management model of a measurement process, and the numbers in Fig. 2.24 refer to ISO 10012: 2003 requirements.

- Recalibration of a measuring instrument is unnecessary if it is within a valid calibration situation. The metrological proof procedure may include mechanisms

[1] The ISO 10012: 2003 standard treats measuring instrument as a measurement equipment. As we did not find this term in VIM, we always replace for instrument.

Still, within requirement *7—Metrological confirmation and realization of measurement processes*, it is essential to highlight the issue of records of the metrological evidence process. These records should include:

- Single description and identification of the instrument: type, model, serial number, manufacturer, etc.
- Date that metrological evidence was performed.
- Evidence results.
- Interval of the following evidence.
- Identification of the procedure (or method, norm, instruction, etc.) of evidence.
- Maximum acceptable or permissible errors.
- Relevant environmental conditions and declaration on necessary corrections.
- Uncertainties involved in calibration.
- Provide details of any intervention (maintenance, adjustment, modification) in the measuring instrument.
- Limitations of use.
- Identification of those who performed the metrological evidence.
- Identification of those who are responsible for any correction of information recorded.
- Single identification of the report or calibration certificate.
- Traceability of measurement results.
- Metrological requirements for intended use.
- Result of calibration performed after, and where required, before any intervention in the measuring instrument.

The standard states that the retention time of metrological evidence records depends on several factors, such as customer requirements, statutory or regulatory requirements, and the manufacturer's civil liability. Records related to measurement standards may need to be kept indefinitely.

2.9.6 Technical Standard and Technical Regulation

2.9.6.1 Technical Standard

ISO (International Organization for Standardization) defines a technical standard as "a document established by consensus and approved by a recognized organism, which provides minimum rules, guidelines or characteristics for activities or their results, aiming at obtaining a great degree of sorting in a given context."

It should be highlighted that technical standards are established by consensus among those interested and approved by a recognized organism. They are also developed for the benefit and cooperation of all interested parties and, in particular, to promote the optimal global economy, taking into account functional conditions and safety requirements.

by DIN (Deutsches Institut Für Normung), in England by BSI (British Standards Institute), and in the United States of America by ANSI (American National Standards Institute).

2.9.6.2 Technical Regulation

A technical regulation is a document a legal authority adopts to do so. It contains mandatory rules and establishes technical requirements, either directly, by reference to technical standards, or by incorporating their content, in whole or in part. In general, technical regulations aim to ensure aspects related to health, safety, environment, consumer protection, and fair competition. Compliance with a technical regulation is mandatory, and non-compliance is illegible and punishable by the corresponding punishment.

Sometimes, a technical regulation establishes the technical rules and requirements for a product, process, or service and can also establish procedures for assessing compliance with regulation, including compulsory certification.

Technical Regulations and International Trade
All countries issue technical regulations. Thus, when it is intended to export a product for a particular market, it is essential to know if the product or service to be exported is subject to a technical regulation in that country in particular.

The WTO Trade Technical Barriers Agreement establishes a series of principles to eliminate unnecessary obstacles to trade, particularly technical barriers related to standards, technical regulations, and compliance assessment procedures that can make it difficult to access products to markets. One of the essential points of the agreement is the understanding that the standards prepared by international standardization bodies (ISO or IEC) constitute the reference for global trade. The agreement stipulates that, whenever possible, governments must adopt technical regulations based on international standards, considering that technical barriers do not constitute those who follow these rules.

Knowing a Little More...
Whenever a government decides to adopt a technical regulation that does not follow an international standard, it should formally notify the other members of the WTO at least 60 days in advance, a justification presented. Other WTO members may request clarification and submit comments and suggestions to the proposed regulation. This information is conveyed by the so-called "Inquiry Points," which are organizations designated by each of the WTO members responsible for notifications of the regulation to be adopted by that country and for receiving notifications made by other countries. Brazil's Inquiry Point is INMETRO.



Fig. 2.25 VIM

2.10 International Vocabulary of Metrology (VIM)

The following text reproduces the introduction of the Bilingual, English, and French editions issued by BIPM and highlights the document's importance (Fig. 2.25).

> *In general, a vocabulary is a "terminological dictionary which contains designations and definitions from one or more specific subject fields" (ISO 1087-1:2000, 3.7.2). The present Vocabulary pertains to metrology, the "science of measurement and its application." It also covers the basic principles governing quantities and units. The field of quantities and units could be treated in many different ways. Clause 1 of this Vocabulary is one such treatment and is based on the principles laid down in the various parts of ISO 31, Quantities and units, currently being replaced by ISO 80000 and IEC 80000 series Quantities and units, and in the SI Brochure, The International System of Units (published by the BIPM).*
>
> *The second edition of the International vocabulary of basic and general terms in metrology (VIM) was published in 1993. The need to cover measurements in chemistry and laboratory medicine for the first time, as well as to incorporate concepts such as those that relate to metrological traceability, measurement uncertainty, and nominal properties, led to this third edition. Its title is now International vocabulary of metrology — Basic and general concepts and associated terms (VIM), to emphasize the primary role of concepts in developing a vocabulary.*

In the GUM, the definitional uncertainty is considered to be negligible concerning the other components of measurement uncertainty. The objective of measurement is to establish a probability that this essentially unique value lies within an interval of measured quantity values based on the information available from measurement.

The IEC scenario focuses on measurements with single readings, permitting the investigation of whether quantities vary in time by demonstrating whether measurement results are compatible. The IEC view also allows non-negligible definitional uncertainties. The validity of the measurement results is highly dependent on the metrological properties of the instrument, as demonstrated by its calibration. The interval of values offered to describe the measurand is the interval of values of measurement standards that would have given the same indications.

In the GUM, the concept of true value is kept for describing the objective of measurement, but the adjective "true" is considered to be redundant. The IEC does not use the concept to describe this objective. In this Vocabulary, the concept and term are retained because of common usage and the importance of the concept.

[VIM—introduction].

2.11 Proposed Exercises

2.11.1 Analyze the following statement: "It is not necessary to calibrate a brand-new instrument from a reputable and traditional manufacturer in the market because the manufacturer guarantees its traceability." Do you agree or disagree? Justify your answer.

2.11.2 What is metrology?

2.11.3 Present some differences between scientific metrology and legal metrology.

2.11.4 What is the function of legal metrology in our society?

2.11.5 According to ISO 10012:2003, present five items should be included in the records of metrological evidence processes.

2.11.6 What do you mean by "influence quantity"?

2.11.7 What is the leading world organism of legal metrology?

2.11.8 What does it mean for a laboratory to be accredited?

2.11.9 What is the difference between technical standards and technical regulation?

2.11.10 What is the importance of the international vocabulary of metrology (VIM)?